

Boundary Conditions Review

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December 8, 2025

Right-Side Dirichlet Boundary Condition

Motivation To maintain second-order accuracy on a staggered grid where the physical boundary lies $\delta = \Delta x/2$ past the last fluid velocity node, we employ the ghost-point method to enforce the Dirichlet condition via Taylor expansion. We present a scheme for the stencil (see Figure 1).

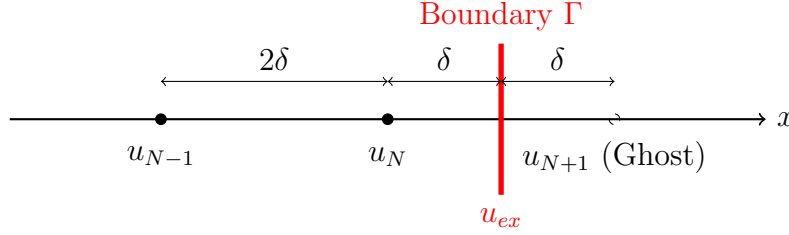


Figure 1: Stencil for right-side tangent boundary condition

Definition of Equations Our goal is to find an expression for u_N'' . This expression cannot be evaluated explicitly, requiring a value u_{N+1} , which lies outside the physical domain. We exploit Taylor expansion, accurate up to second order, for the components of the expression above.

$$\begin{aligned} u_{N+1} &= u_{ex} + u'_{ex}\delta + \frac{1}{2}u''_{ex}\delta^2 + \mathcal{O}(\delta^3) \\ u_N &= u_{ex} + u'_{ex}(-\delta) + \frac{1}{2}u''_{ex}\delta^2 + \mathcal{O}(\delta^3) \\ u_{N-1} &= u_{ex} + u'_{ex}(-3\delta) + \frac{1}{2}u''_{ex}(9\delta^2) + \mathcal{O}(\delta^3) \end{aligned}$$

Symbolic linear system Truncating higher-order terms we can reorganize the three equations above in a linear system for the unknowns u_{N+1} , u'_{ex} , u''_{ex} .

$$\begin{pmatrix} 1 & -\delta & -\frac{1}{2}\delta^2 \\ 0 & -\delta & \frac{1}{2}\delta^2 \\ 0 & -3\delta & \frac{9}{2}\delta^2 \end{pmatrix} \begin{pmatrix} u_{N+1} \\ u'_{ex} \\ u''_{ex} \end{pmatrix} = \begin{pmatrix} u_{ex} \\ u_N - u_{ex} \\ u_{N-1} - u_{ex} \end{pmatrix}$$

The system above gives a second order accurate extrapolation:

$$u_{N+1} = \frac{8}{3}u_{ex} - 2u_N + \frac{1}{3}u_{N-1}$$

Second derivative on the last element Recombining the former result we obtain:

$$u''_N = \frac{2}{3\delta^2}u_{ex} - \frac{1}{\delta^2}u_N + \frac{1}{3\delta^2}u_{N-1}$$

Boundary condition for the linear system Assuming to solve the tridiagonal linear system of dimensions $N \times N$

$$(I - \gamma\partial_{xx})\eta^{n+1} = \xi^{n+1} + \gamma\partial_{xx}\eta^n$$

With the boundary condition

$$\eta^n_\Gamma = u^n_{ex}$$

Then the last row is:

$$\left(1 + \frac{\gamma}{\delta^2}\right)\eta^{n+1}_N - \frac{\gamma}{3\delta^2}\eta^{n+1}_{N-1} = \frac{2\gamma}{3\delta^2}(u^{n+1}_{ex} + u^n_{ex}) + \xi^{n+1}_N - \frac{\gamma}{\delta^2}\eta^n_N + \frac{\gamma}{3\delta^2}\eta^n_{N-1}$$

Expressing in terms of grid spacing

$$\left(1 + \frac{4\gamma}{\Delta x^2}\right)\eta^{n+1}_N - \frac{4\gamma}{3\Delta x^2}\eta^{n+1}_{N-1} = \frac{8\gamma}{3\Delta x^2}(u^{n+1}_{ex} + u^n_{ex}) + \xi^{n+1}_N - \frac{4\gamma}{\Delta x^2}\eta^n_N + \frac{4\gamma}{3\Delta x^2}\eta^n_{N-1}$$

Left-Side Boundary Condition

Motivation While the former formulation for the left boundary exploited the divergence-free condition for the flow field, here we compute another one based only on Taylor expansion as done in the last section. A comparison on numerical results will shortly follow. We present a general scheme for the left boundary (see Figure 2).

Dirichlet condition for normal velocity

Definition of Equations Our goal is to find an expression for u''_0 . This expression cannot be computed explicitly because it requires a value, u_{-1} , which lies outside of the physical domain. We exploit Taylor expansion, accurate up to second order, for the components of the expression above.

Left Boundary Γ

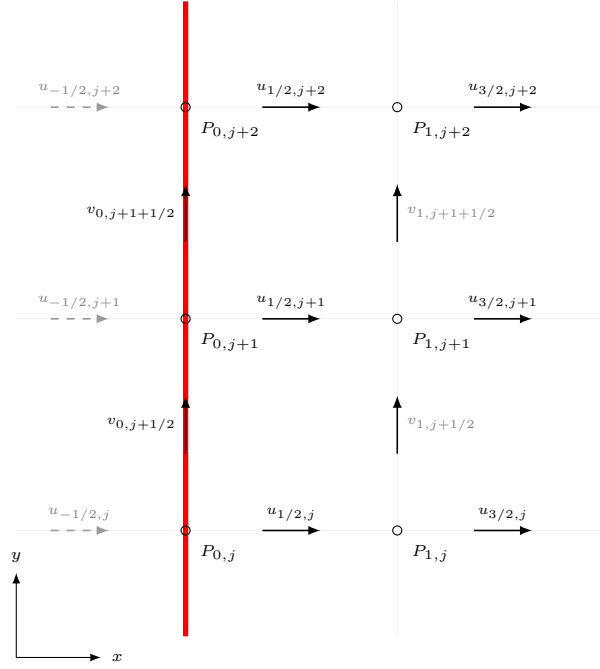


Figure 2: General scheme for left boundary.

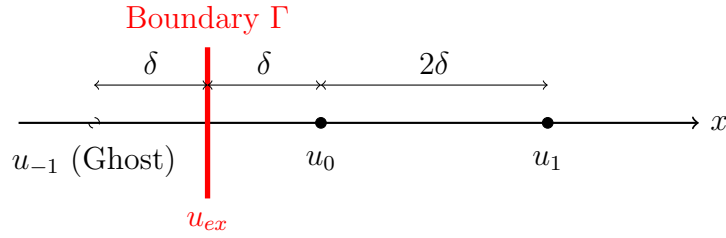


Figure 3: Stencil for left-side normal boundary condition

$$\begin{aligned}
 u_{-1} &= u_{ex} + u'_{ex}(-\delta) + \frac{1}{2}u''_{ex}\delta^2 + \mathcal{O}(\delta^3) \\
 u_0 &= u_{ex} + u'_{ex}\delta + \frac{1}{2}u''_{ex}\delta^2 + \mathcal{O}(\delta^3) \\
 u_1 &= u_{ex} + u'_{ex}(3\delta) + \frac{1}{2}u''_{ex}(9\delta^2) + \mathcal{O}(\delta^3)
 \end{aligned}$$

Symbolic linear system Neglecting second-order infinitesimals we can reorganize the three equations above in a linear system for the unknowns u_{-1} , u'_{ex} , u''_{ex} .

$$\begin{pmatrix} 1 & \delta & -\frac{1}{2}\delta^2 \\ 0 & \delta & \frac{1}{2}\delta^2 \\ 0 & 3\delta & \frac{9}{2}\delta^2 \end{pmatrix} \begin{pmatrix} u_{-1} \\ u'_{ex} \\ u''_{ex} \end{pmatrix} = \begin{pmatrix} u_{ex} \\ u_0 - u_{ex} \\ u_1 - u_{ex} \end{pmatrix}$$

The system above gives a second order accurate extrapolation:

$$u_{-1} = \frac{8}{3}u_{ex} - 2u_0 + \frac{1}{3}u_1$$

Second derivative on the first element Recombining the former result we obtain:

$$u''_0 = \frac{4}{3\Delta x^2}u_1 - \frac{4}{\Delta x^2}u_0 + \frac{8}{3\Delta x^2}u_{ex}$$

Boundary condition for the linear system Recovering the tridiagonal system we are interested in:

$$(I - \gamma\partial_{xx})\eta^{n+1} = \xi^{n+1} + \gamma\partial_{xx}\eta^n$$

With the boundary condition

$$\eta_{\Gamma}^n = u_{ex}^n$$

Then the first row is:

$$\left(1 + \frac{4\gamma}{\Delta x^2}\right)\eta_0^{n+1} - \frac{4\gamma}{3\Delta x^2}\eta_1^{n+1} = \frac{8\gamma}{3\Delta x^2}(u_{ex}^{n+1} + u_{ex}^n) + \xi_0^{n+1} - \frac{4\gamma}{\Delta x^2}\eta_0^n + \frac{4\gamma}{3\Delta x^2}\eta_1^n$$

Left boundary divergence calculation

Once we have solved the symbolic system above, we recall the expression for u'_{ex} , then used in the divergence assembly in the Pressure Step.

$$u'_{ex} = -\frac{1}{3\Delta x}u_1 + \frac{3}{\Delta x}u_0 - \frac{8}{3\Delta x}u_{ex}$$

Numerical results

By analysing only the velocity convergence, we observe that both the divergence-free boundary conditions and the Taylor-based boundary conditions achieve second-order accuracy.

However, the divergence-free formulation exhibits two limitations: it provides only first order spatial convergence to the pressure, and it misrepresents divergence on the boundary.

As the hypothesis of zero divergence on the boundary is not met, the pressure correction step deals with unphysical values. The method enforces the velocity onto a solenoidal field. The velocity, once projected, remains correct, but in an incremental pressure correction scheme this leads to an error on the pressure.

References

- [1] Randall J. LeVeque. *Finite Difference Methods for Ordinary and Partial Differential Equations: Steady-State and Time-Dependent Problems*. Society for Industrial and Applied Mathematics, Philadelphia, 2007.